

Gastrointestinal, Hepatobiliary, and Pancreatic Health

Digestion, absorption, motility, barrier immunity, microbiome, liver, bile, pancreas, nutrition, and gut-brain systems

Volume 03E of the Health Atlas Research Suite

Purpose

Unify digestive physiology, liver and pancreatic function, nutrition, microbiology, symptoms and procedures into a causal and measurement framework that supports safe device and study design.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

Evidence and standards are current through 13 August 2026. This volume does not provide individual endoscopy preparation, antimicrobial regimens, transplant criteria or emergency resuscitation instructions. It does not endorse commercial food-sensitivity, microbiome or intolerance testing without analytical and clinical validation. Human ingestible or catheter devices require formal material, retention, electrical, wireless, privacy and clinical-risk review.

How to use this volume

Digestive health is the ability to ingest, move, digest and absorb food; maintain epithelial and immune tolerance; process nutrients, drugs and toxins; form and circulate bile; produce pancreatic secretions; eliminate waste; and do so without disabling pain, bleeding, infection, malnutrition or loss of participation. The tract is both an organ system and a high-exposure ecosystem coupled to food, microbes, behavior, immunity, metabolism and the brain. A complete profile combines symptom timing and function with nutrition and body stores, stool and blood tests, imaging, endoscopy and histology when indicated, motility and reflux physiology, liver reserve and portal consequences, microbial ecology, medication/exposure history and longitudinal response. A positive molecular or microbiome result is not automatically causal, and a visible lesion is not automatically the source of symptoms.

Questions this volume is designed to answer

1. How do motility, digestion, absorption, barrier immunity, microbiota and gut-brain signaling interact?
2. How should structural disease be distinguished from functional and sensory-motor phenotypes?
3. What do common liver, stool and pancreatic tests measure and fail to measure?
4. When do endoscopy, capsule, imaging, manometry and biopsy add decision value?
5. How can ingestible and microbiome technologies be calibrated and validated?
6. Which preventive and therapeutic interventions address causes versus symptoms?
7. What remains uncertain about microbiome personalization, liver screening and persistent digestive symptoms?

Scope

- Motility, secretion, digestion, absorption, epithelial barrier, immunity and gut-brain sensation.
- Microbiome ecology, nutrition, liver metabolism and clearance, bile, pancreas and portal circulation.
- Luminal, inflammatory, infectious, functional, malabsorptive, hepatic, biliary, pancreatic, bleeding and neoplastic disorders.
- Laboratory/stool testing, endoscopy, capsule systems, imaging, elastography, manometry, pH-impedance, breath tests and pathology.
- Engineering and validation of ingestible, catheter, optical, acoustic, biochemical and microbiome systems.
- Food/public-health prevention, pharmacology, endoscopic/surgical therapy, transplantation, nutrition, rehabilitation and palliation.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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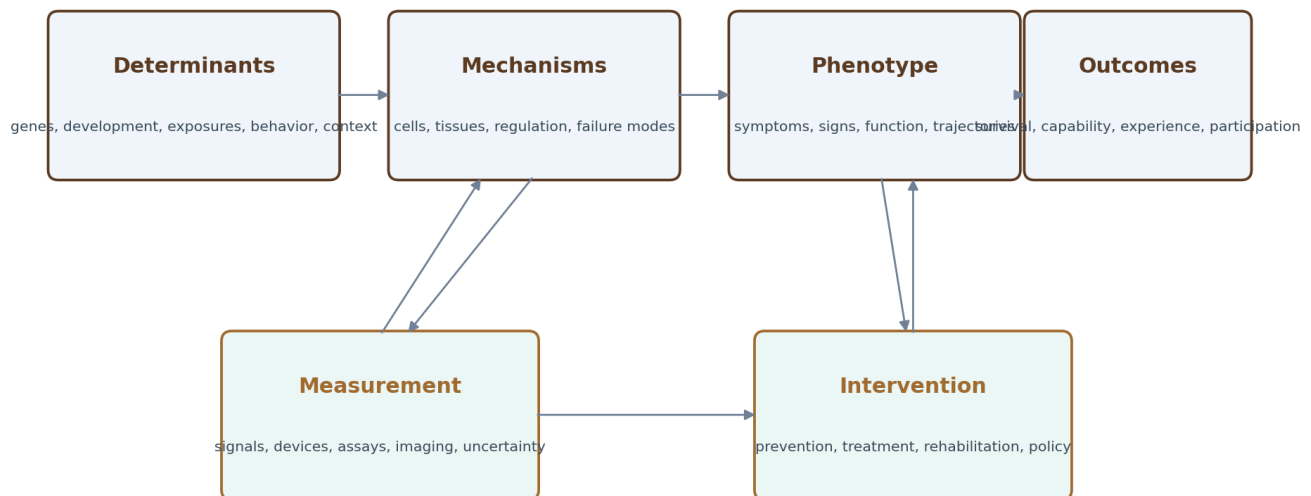
Executive synthesis

Digestive health is the ability to ingest, move, digest and absorb food; maintain epithelial and immune tolerance; process nutrients, drugs and toxins; form and circulate bile; produce pancreatic secretions; eliminate waste; and do so without disabling pain, bleeding, infection, malnutrition or loss of participation. The tract is both an organ system and a high-exposure ecosystem coupled to food, microbes, behavior, immunity, metabolism and the brain. A complete profile combines symptom timing and function with nutrition and body stores, stool and blood tests, imaging, endoscopy and histology when indicated, motility and reflux physiology, liver reserve and portal consequences, microbial ecology, medication/exposure history and longitudinal response. A positive molecular or microbiome result is not automatically causal, and a visible lesion is not automatically the source of symptoms.

Central organizing principle

Unify digestive physiology, liver and pancreatic function, nutrition, microbiology, symptoms and procedures into a causal and measurement framework that supports safe device and study design.

Gastrointestinal, Hepatobiliary, and Pancreatic Health



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.

- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

GI hemorrhage with instability, severe abdominal pain with peritonitis, cholangitis, acute liver failure, severe dehydration, obstruction, perforation or severe pancreatitis are urgent conditions. Ingestible and endoscopic devices can retain, perforate, bleed, infect or interact with imaging; human use requires clinical selection and regulated controls.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to early modern	Diet, stool, vomiting, jaundice, abdominal pain and wasting were central to medical observation.	Established the digestive tract and liver as visible mediators between environment and health.	Mechanisms were framed through humors and food qualities rather than anatomy, microbes or metabolism.
1600s-1800s	Anatomy, secretion experiments, chemical digestion, microscopy and portal-circulation studies developed.	Separated mechanical digestion, enzymes, bile, absorption and hepatic processing.	Cellular transport, endocrine signaling and microbial ecology remained obscure.
1800s	Germ theory, pathology, anesthesia, abdominal surgery and early endoscopy transformed diagnosis.	Linked infection, ulceration, obstruction and tumors to visible lesions and enabled intervention.	Procedures were dangerous, and functional disorders without lesions were marginalized.
1900-1950	Radiography, contrast studies, liver chemistry, transfusion, antibiotics and nutritional science matured.	Made obstruction, bleeding, malnutrition and hepatic failure measurable and treatable.	Tests were nonspecific; viral, immune and molecular causes were poorly separated.
1950-1990	Flexible endoscopy, biopsy, ERCP, ultrasound, CT, parenteral nutrition, acid suppression and hepatitis serology expanded.	Allowed direct visualization, histology, biliary therapy and support during organ failure.	Procedure complications, overuse and medication harms became clear; symptom-lesion mismatch persisted.
1990-2010	H. pylori causality, viral hepatitis molecular testing, transplantation, biologics, capsule endoscopy and high-resolution manometry emerged.	Converted several chronic diseases into treatable etiologies and improved access to small bowel and motility physiology.	Cost, resistance, immune risk and uncertain symptom mechanisms remained.
2010-2026	Elastography, advanced endoscopy, direct-acting antivirals, multi-omics, organoids, microbiome research and metabolic-liver frameworks accelerated.	Enabled noninvasive fibrosis assessment, curative viral therapy and molecular discovery.	Microbiome and AI claims often outpace reproducibility and utility; fatty liver burden reflects food and social systems.

Continued on next page

Period	Development	What it made possible	Residual limitation
Current systems view	Digestive health includes ingestion, motility, digestion, absorption, barrier and immune function, microbiota, liver metabolism/detoxification, bile, pancreas, nutrition and gut-brain experience.	Integrates structural disease with function, symptoms, diet, medications, environment and psychosocial context.	Care remains split among luminal, liver, pancreas, surgery, nutrition and mental-health services.

2.0.1 Diet, stool, vomiting, jaundice, abdominal pain and wasting were central to medical observation.

Context. Digestive health has always linked medicine to food and culture.

Scientific consequence. Established the digestive tract and liver as visible mediators between environment and health.

What remained unresolved. Mechanisms were framed through humors and food qualities rather than anatomy, microbes or metabolism.

Evidence anchors: [5,6]

2.0.2 Anatomy, secretion experiments, chemical digestion, microscopy and portal-circulation studies developed.

Context. Physiology replaced a single stomach-centered model.

Scientific consequence. Separated mechanical digestion, enzymes, bile, absorption and hepatic processing.

What remained unresolved. Cellular transport, endocrine signaling and microbial ecology remained obscure.

Evidence anchors: [3,4]

2.0.3 Germ theory, pathology, anesthesia, abdominal surgery and early endoscopy transformed diagnosis.

Context. Structural disease became the dominant clinical object.

Scientific consequence. Linked infection, ulceration, obstruction and tumors to visible lesions and enabled intervention.

What remained unresolved. Procedures were dangerous, and functional disorders without lesions were marginalized.

Evidence anchors: [5,8]

2.0.4 Radiography, contrast studies, liver chemistry, transfusion, antibiotics and nutritional science matured.

Context. Laboratory and imaging signals became proxies for organ function.

Scientific consequence. Made obstruction, bleeding, malnutrition and hepatic failure measurable and treatable.

What remained unresolved. Tests were nonspecific; viral, immune and molecular causes were poorly separated.

Evidence anchors: [6,27]

2.0.5 Flexible endoscopy, biopsy, ERCP, ultrasound, CT, parenteral nutrition, acid suppression and hepatitis serology expanded.

Context. Gastroenterology became procedure-intensive.

Scientific consequence. Allowed direct visualization, histology, biliary therapy and support during organ failure.

What remained unresolved. Procedure complications, overuse and medication harms became clear; symptom-lesion mismatch persisted.

Evidence anchors: [7,8,9]

2.0.6 H. pylori causality, viral hepatitis molecular testing, transplantation, biologics, capsule endoscopy and high-resolution manometry emerged.

Context. Etiologic treatment and phenotype-specific measurement expanded.

Scientific consequence. Converted several chronic diseases into treatable etiologies and improved access to small bowel and motility physiology.

What remained unresolved. Cost, resistance, immune risk and uncertain symptom mechanisms remained.

Evidence anchors: [7,8,9,11]

2.0.7 Elastography, advanced endoscopy, direct-acting antivirals, multi-omics, organoids, microbiome research and metabolic-liver frameworks accelerated.

Context. The field shifted from organ labels toward interacting metabolic, immune and microbial pathways.

Scientific consequence. Enabled noninvasive fibrosis assessment, curative viral therapy and molecular discovery.

What remained unresolved. Microbiome and AI claims often outpace reproducibility and utility; fatty liver burden reflects food and social systems.

Evidence anchors: [9,13,15]

2.0.8 Digestive health includes ingestion, motility, digestion, absorption, barrier and immune function, microbiota, liver metabolism/detoxification, bile, pancreas, nutrition and gut-brain experience.

Context. A normal endoscopy does not mean normal digestive health, and an abnormal test may not explain symptoms.

Scientific consequence. Integrates structural disease with function, symptoms, diet, medications, environment and psychosocial context.

What remained unresolved. Care remains split among luminal, liver, pancreas, surgery, nutrition and mental-health services.

Evidence anchors: [6,17,23]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Motility and enteric control

Core principle. Coordinated smooth-muscle, sphincter and enteric-neural activity moves and mixes contents while permitting storage and continence.

Control architecture. Enteric circuits, autonomic input, interstitial cells, hormones, stretch, meal composition and central state regulate patterns.

Failure modes. Neuropathy, myopathy, obstruction, inflammation, drugs or maladaptive brain-gut signaling causes dysphagia, reflux, stasis, diarrhea or constipation.

Observable consequences. Swallow studies, transit, manometry, impedance, stool pattern and symptom-meal timing.

Cross-system coupling. Neurologic, endocrine, medication, pelvic-floor and psychosocial systems strongly couple.

Evidence anchors: [3,6,17]

3.0.2 Secretion, digestion and absorption

Core principle. Acid, bicarbonate, bile and enzymes break down food; epithelial transport absorbs water, electrolytes and nutrients.

Control architecture. Neural/hormonal signals, pancreatic and hepatic function, mucosal surface and transporters match secretion to meals.

Failure modes. Enzyme deficiency, mucosal injury, transporter defects, short bowel, bile deficiency or bacterial overgrowth causes maldigestion/malabsorption.

Observable consequences. Nutrient labs, stool fat/elastase, breath tests, endoscopy/biopsy, imaging and growth/weight/function.

Cross-system coupling. Nutrition, endocrine metabolism, microbiome, liver, pancreas and kidney status alter phenotype.

Evidence anchors: [3,4,6]

3.0.3 Barrier and mucosal immunity

Core principle. A single-cell epithelial barrier samples food and microbes while preventing invasion and excessive inflammation.

Control architecture. Tight junctions, mucus, antimicrobial products, secretory antibodies, immune tolerance and epithelial renewal maintain balance.

Failure modes. Infection, immune dysregulation, ischemia, drugs or genetic defects cause inflammation, permeability and ulceration.

Observable consequences. Endoscopy/biopsy, inflammatory markers, microbiology, permeability research assays and clinical function.

Cross-system coupling. Immune, infection, nutrition and microbiome domains intersect.

Evidence anchors: [5,7,10]

3.0.4 Gut microbiome ecology

Core principle. Microbes transform nutrients and bile acids, resist pathogens and interact with immune, metabolic and neural systems.

Control architecture. Diet, host secretions, transit, anatomy, medications, infection and ecological competition shape communities.

Failure modes. Antibiotics, altered anatomy, inflammation, diet or immune failure can change functions, but dysbiosis is often a description rather than a cause.

Observable consequences. Culture/PCR for pathogens, metagenomics/metabolomics, ecological diversity and functional assays in research.

Cross-system coupling. Infection, immunity, environment, pharmacology and psychosocial behavior all influence sampling and effect.

Evidence anchors: [10,11,13]

3.0.5 Hepatic metabolism and clearance

Core principle. The liver processes nutrients, synthesizes proteins, stores substrates, transforms drugs/toxins and excretes bile products.

Control architecture. Portal flow, hepatocyte zones, endocrine signals, enzymes, transporters and immune cells coordinate flux.

Failure modes. Steatosis, inflammation, cholestasis, necrosis, fibrosis, vascular obstruction or failure impairs synthesis, clearance and homeostasis.

Observable consequences. Chemistries, synthetic function, elastography, imaging, pathology and portal/hemodynamic consequences.

Cross-system coupling. Cardiovascular, metabolic, hematologic, immune, infectious and drug systems converge.

Evidence anchors: [3,5,9]

3.0.6 Bile formation and enterohepatic circulation

Core principle. Bile acids emulsify lipids and act as signaling molecules; transport cycles among liver, bile ducts, intestine and portal blood.

Control architecture. Transporters, gallbladder, microbiota, meal signals and hepatic synthesis regulate pool and flow.

Failure modes. Stones, obstruction, transporter/immune disease, infection or drug injury causes pain, cholestasis, malabsorption and inflammation.

Observable consequences. Liver tests, ultrasound/MRCP, endoscopy, bile-acid studies and pathology.

Cross-system coupling. Microbiome, endocrine metabolism, pancreas and pharmacology interact.

Evidence anchors: [3,8,9]

3.0.7 Pancreatic exocrine and endocrine integration

Core principle. Acinar enzymes and ductal bicarbonate enable digestion while islet hormones regulate metabolism.

Control architecture. Meal neural/hormonal signals, duct flow and protective enzyme activation control secretion.

Failure modes. Premature enzyme activation, obstruction, fibrosis, tumor or genetic disease causes pancreatitis and insufficiency; islet injury causes diabetes.

Observable consequences. Lipase in context, imaging, elastase, nutrition, endoscopic ultrasound and glucose phenotype.

Cross-system coupling. Endocrine, biliary, alcohol/toxin, genetic and nutritional systems couple.

Evidence anchors: [6,8,14]

3.0.8 Splanchnic circulation and portal pressure

Core principle. The gut and spleen drain through the portal system; hepatic resistance and collateral pathways determine pressure.

Control architecture. Blood flow, sinusoidal tone, fibrosis, thrombosis, cardiac pressure and vasoactive signals regulate the circuit.

Failure modes. Cirrhosis or vascular obstruction causes portal hypertension, varices, ascites, splenomegaly and gut permeability changes.

Observable consequences. Imaging, platelets/spleen, elastography, endoscopy, pressure gradients selectively and decompensation phenotype.

Cross-system coupling. Cardiovascular, renal, hematologic and immune systems determine consequences.

Evidence anchors: [2,9,16]

3.0.9 Visceral sensation and gut-brain interaction

Core principle. Mechanical, chemical, immune and microbial signals reach enteric, spinal and vagal pathways; central prediction and attention shape symptoms.

Control architecture. Descending modulation, stress systems, learning, sleep and local inflammation regulate perception and behavior.

Failure modes. Sensitization, dysmotility, hypervigilance or central-autonomic dysregulation can produce severe symptoms without destructive disease.

Observable consequences. Symptom pattern, function, dietary context, validated criteria, targeted exclusion and psychophysiologic measures in research.

Cross-system coupling. Mental health, pain, autonomic and social domains are integral and should not imply symptoms are unreal.

Evidence anchors: [17,18,19]

3.0.10 Nutrition and body stores

Core principle. Digestive function must ultimately support energy, protein, micronutrients, hydration, growth and repair.

Control architecture. Appetite, food access, chewing/swallowing, absorption, storage, inflammation and energy demand determine status.

Failure modes. Food insecurity, eating disorder, malabsorption, inflammation, cancer, surgery or organ failure causes deficiency or excess.

Observable consequences. Growth/weight trajectory, diet/access, exam, functional measures and targeted labs rather than one biomarker.

Cross-system coupling. Oral, metabolic, immune, musculoskeletal, psychosocial and population systems are coupled.

Evidence anchors: [20,21,22]

3.0.11 Repair, fibrosis and carcinogenesis

Core principle. Rapidly renewing epithelium and hepatic/pancreatic repair respond to recurrent injury; chronic maladaptive repair can produce fibrosis or neoplasia.

Control architecture. Stem/progenitor cells, extracellular matrix, immune resolution, microbiota and genomic surveillance regulate recovery.

Failure modes. Persistent infection, inflammation, toxins, metabolic injury or mutations drive scarring, dysplasia and cancer.

Observable consequences. Endoscopy/biopsy, elastography, imaging, molecular/pathologic classification and longitudinal surveillance.

Cross-system coupling. Cross-system inflammation, aging, infection and environment shape risk.

Evidence anchors: [5,24,25]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Henderson-Hasselbalch for gastric/physiologic buffers	$pH = pK_a + \log \frac{[A^-]}{[HA]}$	Relates pH to a weak acid/base pair and is useful for acid-base and ionization reasoning.	GI compartments are dynamic, buffered and nonideal; local pH and activity differ from simple solutions.
Hepatic extraction	$E = \frac{C_{in} - C_{out}}{C_{in}}$	Fraction removed across the liver depends on flow, protein binding and intrinsic clearance.	Requires defined sampling and steady conditions; shunting and zonation complicate interpretation.
Portal pressure gradient	$\Delta P = Q R$	Pressure rise reflects portal flow and vascular resistance.	Collateral flow and nonlinearity make indirect markers incomplete; clinical thresholds are context-specific.
Transit velocity	$v = \Delta x / \Delta t$	A simple summary of movement through a defined segment.	GI transit is intermittent, segmental and meal/state dependent; one average can hide retention and bursts.
Nutrient balance	$\Delta S = I - A - U - L$	Store change reflects intake minus unabsorbed, utilized and lost fractions.	Each term is difficult to measure and adapts over time; body composition is multicompartamental.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within gastrointestinal, hepatobiliary, and pancreatic health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within gastrointestinal, hepatobiliary, and pancreatic health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[15]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Evidence anchors: ^[22]

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[19]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: ^[19,20]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: [20,48]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: [39]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the digestive syndrome, structural lesion, functional phenotype, or organ-failure state, but cross-system processes and functional effects are retained.

5.0.1 Gastroesophageal reflux disease

Definition and phenotype. Troublesome symptoms or complications from reflux of gastric contents into esophagus or beyond.

Pathophysiology. Transient sphincter relaxation, pressure gradients, hiatal anatomy, clearance and mucosal sensitivity interact.

Determinants and modifiers. Obesity, pregnancy, meals, medications, tobacco, anatomic factors and sleep position modify expression.

Assessment. Symptom pattern and alarm features; empiric approach in selected cases; endoscopy for complications/alarm; pH-impedance when objective correlation changes management; manometry before selected procedures.

Intervention logic. Behavioral/mechanical measures, acid suppression, alginate or other medication, and surgery/endoscopic procedures for carefully selected objective disease.

Known limits and open questions. Symptoms correlate imperfectly with acid events; long-term medication/procedure benefit and harms depend on phenotype.

Suite links. 03D obesity; 03N pregnancy; 05 brain-gut; 14 pharmacology/procedure.

Evidence anchors: [6,7,8]

5.0.2 Peptic ulcer disease and *Helicobacter pylori*

Definition and phenotype. Mucosal ulceration in stomach or duodenum commonly caused by *H. pylori* or NSAID-related injury.

Pathophysiology. Acid-peptic injury exceeds mucosal defense through infection, prostaglandin suppression, ischemia or hypersecretory states.

Determinants and modifiers. *H. pylori* exposure, NSAIDs/antithrombotics, tobacco, critical illness and rare gastrinoma.

Assessment. Endoscopy when bleeding/alarm/age-risk requires; noninvasive or biopsy *H. pylori* testing with attention to medication effects; assess anemia and complications.

Intervention logic. *H. pylori* eradication with resistance-aware regimen, stop/reduce injury drugs when possible, acid suppression and endoscopic/surgical treatment of complications.

Known limits and open questions. Resistance and reinfection vary; test-of-cure is essential; dyspepsia may persist after ulcer healing.

Suite links. 08 infection/AMR; 03G bleeding; 14 drugs.

Evidence anchors: [7,8,11]

5.0.3 Functional dyspepsia and disorders of gut-brain interaction

Definition and phenotype. Chronic digestive symptoms defined by positive symptom patterns after appropriate evaluation, without a structural disease sufficient to explain them.

Pathophysiology. Visceral hypersensitivity, motility/accommodation changes, mucosal/immune signals, microbiome, central prediction, stress and learning interact.

Determinants and modifiers. Infection history, anxiety/depression, sleep, diet, medication, trauma/stress and social context may modify but are not required.

Assessment. Positive criteria, alarm review, focused testing by probability, *H. pylori* where appropriate, symptom/function and diet/medication assessment.

Intervention logic. Explanation and therapeutic alliance, meal/diet strategies, acid or motility treatment, neuromodulators and psychological therapies according to phenotype.

Known limits and open questions. No single biomarker; broad diets and microbiome tests can overmedicalize; placebo/context response is part of treatment, not evidence of unreality.

Suite links. 05 cognition/emotion; 06 psychosocial; 14 behavior/pharmacology.

Evidence anchors: [7,17,18]

5.0.4 Irritable bowel syndrome

Definition and phenotype. Recurrent abdominal pain related to defecation and stool change, classified by positive symptom criteria after appropriate evaluation.

Pathophysiology. Motility, visceral sensation, barrier/immune signaling, microbiome, bile acids and central-autonomic processing contribute variably.

Determinants and modifiers. Post-infectious state, sex/life stage, diet, stress, sleep, medications and social factors modify expression.

Assessment. Positive symptom/stool phenotype, alarm and family history, limited targeted labs such as celiac/inflammation testing by context, pelvic-floor assessment if indicated.

Intervention logic. Education, soluble fiber or diet trial with nutrition safeguards, bowel-pattern therapy, neuromodulators and gut-directed psychological therapy; avoid repeated low-yield testing.

Known limits and open questions. Mechanistic subtypes are not yet routinely actionable; restrictive diets risk nutritional and social harm.

Suite links. 05 brain-gut; 03D nutrition; 06 psychosocial; 10 function.

Evidence anchors: [6,7,32]

5.0.5 Inflammatory bowel disease

Definition and phenotype. Chronic immune-mediated Crohn disease or ulcerative colitis causes relapsing intestinal inflammation and extraintestinal effects.

Pathophysiology. Barrier dysfunction, microbial interaction and dysregulated immunity in genetically susceptible hosts produce ulceration, fibrosis, fistula or colitis.

Determinants and modifiers. Genetics, smoking, prior infection/drugs, diet/environment correlates and health-system access influence course.

Assessment. Symptoms and labs, fecal inflammation, ileocolonoscopy/biopsy, cross-sectional imaging, infection exclusion and objective activity/complication assessment.

Intervention logic. Nutrition and vaccination, anti-inflammatory/immune/biologic or small-molecule therapy, surgery for complications or refractory disease, cancer surveillance and psychosocial care.

Known limits and open questions. Symptoms can persist without active inflammation; biomarkers and therapeutic drug levels have context-specific utility; long-term sequencing remains uncertain.

Suite links. 03H immunity; 08 infection; 14 immunotherapy; 10 function.

Evidence anchors: [7,8,41]

5.0.6 Celiac disease and immune-mediated enteropathy

Definition and phenotype. Immune-mediated small-intestinal injury triggered by gluten in genetically susceptible people, with intestinal or extraintestinal phenotypes.

Pathophysiology. Dietary antigen presentation and adaptive immunity cause villous injury and malabsorption.

Determinants and modifiers. Genetic susceptibility, exposure and immune context; family/autoimmune association.

Assessment. Serology while consuming gluten with total IgA context, endoscopic biopsy according to age/guideline and diagnosis, nutritional and bone assessment; avoid diagnosis from commercial sensitivity panels.

Intervention logic. Lifelong gluten-free diet with expert nutrition support, correction of deficiency, monitoring and evaluation of nonresponse.

Known limits and open questions. Serology/biopsy discordance and refractory disease require expertise; dietary burden and contamination are major outcomes.

Suite links. 03H immunity; 03D bone/metabolism; 06 social burden.

Evidence anchors: [6,7]

5.0.7 Malabsorption and short bowel syndromes

Definition and phenotype. Insufficient digestion or absorptive surface causes nutrient, fluid and electrolyte deficits.

Pathophysiology. Mucosal loss, enzyme/bile deficiency, altered transit, bacterial overgrowth or resection reduces effective absorption and adaptation.

Determinants and modifiers. Celiac/IBD, pancreatic/liver disease, surgery, infection, radiation, congenital transport defects and medications.

Assessment. Diet/stool pattern, weight/growth, targeted nutrients, stool/breath/endoscopic tests and anatomy; quantify hydration and kidney/liver complications.

Intervention logic. Cause treatment, individualized diet and oral/enteral/parenteral replacement, antimotility/secretory therapy, intestinal rehabilitation and selected trophic agents/transplant.

Known limits and open questions. Tests have limited specificity; home parenteral support carries infection/liver burden; adaptation is individual.

Suite links. 03D nutrition; 03F kidney/electrolyte; 08 infection; 10 function.

Evidence anchors: [6,30,31]

5.0.8 Constipation and defecatory disorders

Definition and phenotype. Infrequent, difficult or incomplete defecation from transit, pelvic-floor, structural, medication or systemic causes.

Pathophysiology. Colonic propulsion, stool water, rectal sensation and coordinated pelvic-floor relaxation can fail separately or together.

Determinants and modifiers. Diet/fluids, activity, medications, neurologic/endocrine disease, pregnancy, aging, pain and toileting access.

Assessment. Alarm and medication/systemic review, rectal examination; targeted colon evaluation; transit and anorectal testing for refractory or suspected outlet dysfunction.

Intervention logic. Fiber/osmotic or other laxative strategy by phenotype, medication changes, pelvic-floor biofeedback for dyssynergia and selected procedures.

Known limits and open questions. Stool frequency alone is incomplete; consumer transit devices and microbiome claims require validation.

Suite links. 03C autonomic; 03F pelvic/urinary; 03N pregnancy; 10 rehabilitation.

Evidence anchors: [6,7]

5.0.9 Acute and chronic diarrhea

Definition and phenotype. Increased stool water/frequency from infection, inflammation, osmotic load, secretion, malabsorption, drugs or altered transit.

Pathophysiology. Epithelial secretion/absorption imbalance and motility respond to pathogens, toxins, bile acids, immune injury and nutrients.

Determinants and modifiers. Food/water exposure, travel, antibiotics, immune state, medications, surgery, diet and endemic ecology.

Assessment. Duration/severity, hydration and exposure; selective stool microbiology/inflammation, labs, endoscopy or malabsorption testing based on risk; avoid indiscriminate panels.

Intervention logic. Rehydration and electrolyte replacement, cause-specific therapy, infection control and nutrition; antimicrobials only when indicated.

Known limits and open questions. Multiplex detection may find colonization; post-infectious symptoms and chronic unexplained diarrhea require mechanism-based evaluation.

Suite links. 08 infection/AMR; 03F volume/electrolyte; 11 sanitation.

Evidence anchors: [6,7,11]

5.0.10 Gastrointestinal bleeding

Definition and phenotype. Acute or chronic blood loss from upper, small-bowel, colonic or anorectal sources.

Pathophysiology. Mucosal injury, portal hypertension, vascular lesions, tumor; inflammation or coagulation/medication effects causes hemorrhage.

Determinants and modifiers. Ulcer/H. pylori, liver disease, antithrombotics/NSAIDs, age, malignancy, vascular disease and inherited bleeding.

Assessment. Hemodynamic and blood-count assessment, medication/coagulation review; endoscopy/colonoscopy, CT angiography, capsule or enteroscopy by presentation and stability.

Intervention logic. Resuscitation and blood strategy, medication reversal selectively, acid/vasoactive treatment by cause, endoscopic/radiologic/surgical hemostasis and prevention of recurrence.

Known limits and open questions. Hemoglobin lags acute loss; risk scores do not replace clinical state; transfusion and timing must be individualized.

Suite links. 03G hematology; 03A hemodynamics; 12 emergency systems.

Evidence anchors: [8,16,27]

5.0.11 Metabolic dysfunction-associated steatotic liver disease

Definition and phenotype. Hepatic steatosis associated with metabolic risk, spanning simple steatosis through inflammation, fibrosis and cirrhosis.

Pathophysiology. Insulin resistance, adipose flux, de novo lipogenesis, lipotoxicity, inflammation and fibrogenesis interact with genetic susceptibility.

Determinants and modifiers. Adiposity, diabetes, dyslipidemia, sleep apnea, diet/activity environment, medications and genetic variants.

Assessment. Identify steatosis and metabolic risks; exclude important alternatives; use noninvasive fibrosis scores/elastography and specialist evaluation for high-risk/discordant cases.

Intervention logic. Weight and metabolic treatment, cardiovascular risk reduction, alcohol/medication review, selected liver-directed drugs by stage, cirrhosis surveillance and transplant care.

Known limits and open questions. Steatosis amount is not fibrosis; nomenclature and drug evidence are evolving; population screening strategy remains debated.

Suite links. 03D metabolic; 03A cardiovascular; 03B sleep; 14 therapeutics.

Evidence anchors: [9,14,15]

5.0.12 Alcohol-associated liver and pancreatic disease

Definition and phenotype. Alcohol exposure can cause steatosis, hepatitis, fibrosis/cirrhosis, pancreatitis, cancer risk and nutritional/neurologic harm.

Pathophysiology. Dose-pattern and susceptibility drive oxidative injury, inflammation, fibrosis and pancreatic acinar toxicity.

Determinants and modifiers. Quantity/pattern, genetics, sex/body composition, coexisting metabolic or viral disease, nutrition and social context.

Assessment. Nonjudgmental exposure assessment, organ and nutrition testing, withdrawal risk, fibrosis evaluation and alternative causes.

Intervention logic. Alcohol-use treatment and social support, nutrition, management of withdrawal and organ complications, selected severe hepatitis therapy and transplant pathways with equitable criteria.

Known limits and open questions. No safe threshold is identical for every outcome; underreporting and stigma impair care; biomarkers cannot replace conversation.

Suite links. 05 substance behavior; 06 social; 14 addiction treatment; 15 stigma/equity.

Evidence anchors: [9,48]

5.0.13 Viral hepatitis

Definition and phenotype. Hepatitis viruses cause acute or chronic inflammation, fibrosis, cirrhosis, liver failure and cancer with distinct transmission and treatment.

Pathophysiology. Viral replication and host immunity produce direct/immune injury; chronic infection enables fibrosis and carcinogenesis.

Determinants and modifiers. Blood/sexual/perinatal/fecal-oral exposure depending virus, vaccination, injection safety, migration, incarceration and access.

Assessment. Serology and nucleic-acid tests interpreted by virus/stage; liver severity and coinfection evaluation; screening by risk/population guidance.

Intervention logic. Vaccination where available, curative antivirals for hepatitis C, suppressive therapy for hepatitis B, prevention of transmission and surveillance for advanced disease.

Known limits and open questions. Cure/suppression does not erase advanced fibrosis risk; elimination depends on testing and delivery systems, not drugs alone.

Suite links. 08 infection; 11 public health; 03N perinatal; 15 equity.

Evidence anchors: [9,28,30]

5.0.14 Cirrhosis and portal-hypertension decompensation

Definition and phenotype. Advanced diffuse fibrosis and nodular remodeling causes portal hypertension and impaired hepatic function.

Pathophysiology. Increased intrahepatic resistance and vasodilation produce varices, ascites, kidney dysfunction, encephalopathy, infection vulnerability and cancer risk.

Determinants and modifiers. Any chronic liver injury, continued exposure, metabolic disease and access to etiologic treatment determine progression.

Assessment. Clinical/laboratory synthetic function, fibrosis/portal markers, imaging, endoscopy selectively, ascites analysis, encephalopathy and transplant evaluation.

Intervention logic. Treat cause; prevent/manage variceal bleeding, ascites, encephalopathy, infection and malnutrition; surveillance and transplant referral; palliative care when appropriate.

Known limits and open questions. Compensated disease is heterogeneous; prognostic scores omit frailty and social access; procedure benefits vary by physiology.

Suite links. 03F kidney; 03A circulation; 03G hemostasis; 10 function/palliative.

Evidence anchors: [9,30,33]

5.0.15 Gallstone and biliary disease

Definition and phenotype. Stones, inflammation, infection, strictures or tumors obstruct bile flow and cause pain, jaundice or pancreatitis.

Pathophysiology. Supersaturation and gallbladder stasis form stones; obstruction raises pressure and permits inflammation/infection.

Determinants and modifiers. Age, sex hormones/pregnancy, obesity/rapid weight loss, hemolysis, ileal disease, genetics and anatomy.

Assessment. Pain/sepsis pattern, liver tests and ultrasound; MRCP/EUS for duct uncertainty; ERCP mainly therapeutic; evaluate malignancy when appropriate.

Intervention logic. Analgesia and antibiotics for infection, cholecystectomy for indicated symptomatic/complicated disease, endoscopic drainage/stone removal and cause-specific stricture care.

Known limits and open questions. Incidental stones usually do not explain nonspecific symptoms; imaging sensitivity and procedure risk require selection.

Suite links. 03D metabolism; 03N pregnancy; 14 procedure; 08 infection.

Evidence anchors: [8,9,11]

5.0.16 Acute and chronic pancreatitis

Definition and phenotype. Acute autodigestive inflammation or chronic fibroinflammatory destruction causes pain, local/systemic complications, exocrine insufficiency and diabetes.

Pathophysiology. Premature enzyme activation, duct obstruction, toxic/metabolic/genetic injury and inflammatory amplification damage acinar/duct tissue.

Determinants and modifiers. Gallstones, alcohol, triglycerides, medications, trauma/procedure, smoking, genetics and autoimmune disease.

Assessment. Characteristic pain plus enzymes and/or imaging for acute disease; severity/organ failure; chronic imaging/function, cause and nutrition/diabetes assessment.

Intervention logic. Supportive acute care and cause treatment; drainage/procedure for selected complications; abstinence/smoking cessation, enzymes/nutrition, pain strategy and diabetes management chronically.

Known limits and open questions. Enzyme level does not grade severity; chronic pain mechanism and optimal intervention sequence are uncertain.

Suite links. 03D diabetes/lipids; 04 systemic inflammation; 14 procedures/pain.

Evidence anchors: [6,7,8]

5.0.17 Digestive and hepatopancreatobiliary cancers

Definition and phenotype. Malignancies of esophagus, stomach, intestine, liver, bile ducts or pancreas have distinct precursor, molecular and stage pathways.

Pathophysiology. Chronic injury/infection/inflammation, inherited susceptibility and somatic evolution drive dysplasia, invasion and metastasis.

Determinants and modifiers. Tobacco/alcohol, obesity/metabolic disease, H. pylori, viral hepatitis, IBD, diet patterns, age and inherited syndromes.

Assessment. Risk-based screening/surveillance where evidence supports; endoscopy/imaging, tissue and molecular staging, nutrition and functional reserve.

Intervention logic. Prevention and etiologic treatment, surgery, endoscopic therapy, radiation and systemic/targeted/immune therapy by type/stage; palliative support.

Known limits and open questions. Screening is beneficial only in defined populations; incidental lesions, overdiagnosis and resistance remain.

Suite links. 04 cancer; 08 infection; 11 screening; 14 oncology.

Evidence anchors: [8,9,24,26]

A safe digestive measurement platform

GI data combine episodic symptoms, ingestible/catheter sensors, imaging, specimens and behavioral exposures. Time alignment with meals, medications, sleep, posture and bowel events is essential.

6.1 Core noninvasive platform

- Structured meal, symptom, stool-form, medication and exposure diary with timestamps.
- Weight and hydration trajectory; selected home vital signs.
- Validated stool specimen collection with chain of custody and cold/storage metadata.
- Wearable activity/sleep only as context, not a diagnosis.
- Environmental/food exposure records where infection or intolerance is studied.
- Optional ingestible device only after retention and safety analysis.
- Common data schema linking procedures, pathology, laboratory, imaging and interventions.

Selected evidence anchors: [6,43,44]

6.2 Ingestible device hazards

- Retention at stricture, obstruction, aspiration or delayed passage.
- Battery, heat, material and leakage safety.
- Wireless failure, localization error and lost data.
- MRI incompatibility or unrecognized retained capsule.
- Misinterpretation of pH, pressure, gas or transit without anatomy and context.
- Privacy implications of continuous internal-state data.

Selected evidence anchors: [35,36,38]

From liver tests to liver phenotype

Common serum tests do not directly measure the same thing: enzymes suggest injury, cholestatic markers suggest impaired flow, bilirubin reflects production/conjugation/excretion, while albumin/INR are influenced by synthesis and many nonhepatic factors.

7.1 Interpretive layers

Layer	Examples	Question answered
Injury pattern	ALT/AST and context.	Is hepatocellular injury plausible and how is it trending?
Cholestasis	Alkaline phosphatase, GGT, bilirubin, imaging.	Is bile formation/flow impaired and where?
Synthetic/system consequence	INR, albumin, glucose, ammonia in context.	Is organ reserve or systemic homeostasis impaired?
Fibrosis/portal risk	Scores, elastography, platelets/spleen, imaging.	Is advanced scarring/portal hypertension likely?
Etiology	Exposure, metabolic, viral, immune, genetic and vascular tests.	What causal pathway should be treated?

Selected evidence anchors: [9,45]

Microbiome claim ladder

A responsible microbiome program distinguishes detection, association, mechanism and intervention. Skipping levels produces attractive but unreliable personalized recommendations.

8.1 Evidence ladder

- Analytical validity: the platform reproducibly detects the claimed organisms/functions.
- Ecological description: the feature differs across defined groups after controlling batch and major covariates.
- Temporal prediction: the feature precedes outcome in external cohorts.
- Mechanistic support: perturbation changes a relevant pathway in models or humans.
- Clinical validity: the signature adds calibrated information beyond simple variables.
- Clinical utility: acting on it improves outcomes without excess harm or cost.

Selected evidence anchors: [13,42,45]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current gastrointestinal, hepatobiliary, and pancreatic health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

9.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current gastrointestinal, hepatobiliary, and pancreatic health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

Evidence anchors: ^[45]

9.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[23]

9.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[3,5]

9.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

Evidence anchors: ^[45]

9.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[23]

9.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Evidence anchors: ^[43,44]

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

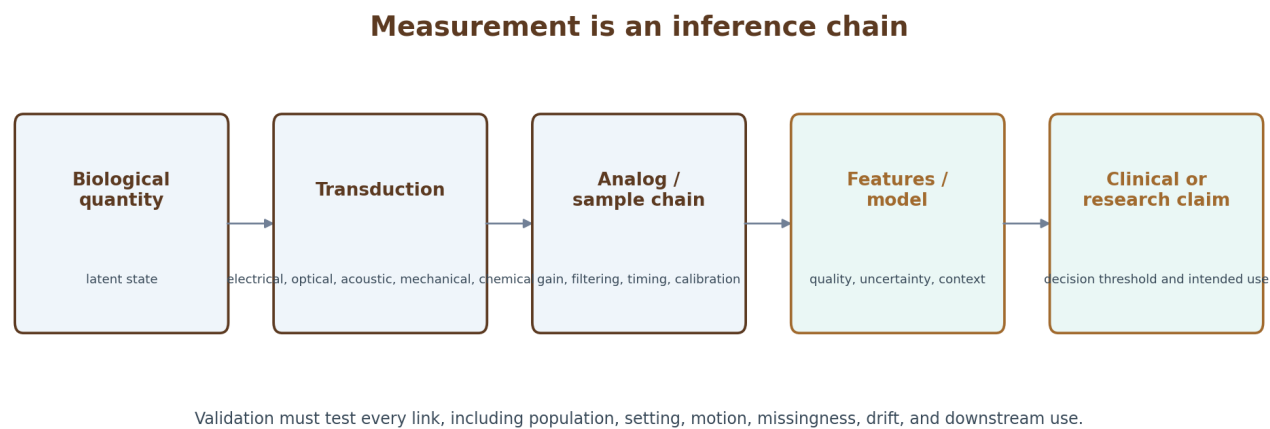


Figure 10.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
History, abdominal and nutritional examination	Noninvasive	Symptoms, meal/stool/exposure/medication history, inspection, palpation, auscultation, rectal/oral and nutrition/function assessment.	Urgency, syndrome, localization, alarm features and test strategy.	High value and low burden.	Symptoms are overlapping; examination sensitivity is limited; stigma and recall affect diet/alcohol/stool reports.
Blood chemistry, hematology and synthetic markers	Minimally invasive specimen	Enzymatic, immunologic and coagulation assays.	Anemia/inflammation, cholestatic/hepatocellular pattern, albumin/INR context, electrolytes and organ effects.	Widely available and longitudinal.	Markers are nonspecific and influenced by muscle, nutrition, anticoagulants, hemolysis and acute illness; normal values may coexist with advanced disease.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Stool microbiology and molecular panels	Noninvasive specimen	Culture, antigen, microscopy or nucleic-acid detection of pathogens/toxins.	Evidence of selected infectious agents and resistance markers.	Can rapidly identify treatable and outbreak-relevant causes.	Colonization, prolonged shedding, broad-panel incidental detection and collection/preanalytic errors.
Fecal inflammatory, pancreatic and occult-blood tests	Noninvasive specimen	Immunochemical or enzymatic markers such as calprotectin, elastase or hemoglobin.	Intestinal inflammation proxy, exocrine insufficiency proxy or screening signal.	Low-burden triage and monitoring for defined claims.	Medication, dilution, bleeding source, age and pretest probability; not diagnostic alone.
Upper endoscopy and colonoscopy	Invasive optical	Flexible video endoscopy with biopsy, polypectomy, hemostasis and therapy.	Mucosal anatomy, lesions, histology and interventions.	Direct visualization and combined diagnosis/treatment.	Sedation, perforation, bleeding, missed lesions, preparation quality, interval cancer and overuse.
Capsule endoscopy and ingestible imaging	Minimally invasive ingestible	Swallowed cameras or sensors transmit images/physiology through GI tract.	Small-bowel mucosa, bleeding source and selected transit/pH/temperature data.	Accesses regions beyond standard endoscopy without sedation.	Retention, incomplete study, localization, image burden, false positives and inability to biopsy/treat.
Endoscopic ultrasound	Invasive acoustic/endoscopic	High-frequency ultrasound from GI lumen with needle sampling or therapy.	Pancreatic/biliary/subepithelial lesions, staging and drainage access.	High-resolution and tissue acquisition.	Sedation, bleeding, infection, pancreatitis, sampling error and operator dependence.
ERCP and therapeutic biliary/pancreatic endoscopy	Invasive fluoroscopic/endoscopic	Cannulation, contrast fluoroscopy, sphincterotomy, extraction, dilation, stenting and sampling.	Duct diagnosis during therapy, drainage and obstruction treatment.	Powerful minimally invasive therapy.	Pancreatitis, bleeding, infection, perforation and radiation; diagnostic-only use is usually inappropriate when safer imaging exists.
Abdominal ultrasound and Doppler	Noninvasive acoustic	Reflected sound and frequency shift image organs, fluid and vessels.	Gallstones, ducts, liver texture, masses, ascites, portal/hepatic flow and procedure guidance.	Portable, inexpensive and no ionizing radiation.	Body habitus/bowel gas, operator and protocol dependence; texture is nonspecific.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Elastography	Noninvasive to minimally invasive	Ultrasound or MR measures tissue response to shear waves as a stiffness proxy.	Liver stiffness and sometimes fat/other tissue properties.	Noninvasive fibrosis-risk stratification and longitudinal potential.	Inflammation, congestion, cholestasis, meals, obesity and platform differences alter stiffness; not direct histology.
CT, MRI and MRCP	Noninvasive/minimally invasive imaging	Minimally-invasive, sectional X-ray or magnetic imaging with contrast and duct-focused sequences.	Inflammation, obstruction, masses, vasculature, staging, complications and bile/pancreatic ducts.	Comprehensive anatomy and rapid acute assessment with CT; high soft-tissue/duct contrast with MRI.	Radiation/contrast, motion, incidental findings, protocol and reconstruction dependence.
pH-impedance monitoring	Minimally invasive catheter or wireless	Electrical impedance and pH detect bolus movement and acid exposure.	Reflux burden, acid/nonacid events and symptom association.	Objective physiology when diagnosis or procedure selection is uncertain.	Catheter behavior change, day variability, symptom-report bias, thresholds and association statistics.
High-resolution manometry	Minimally invasive catheter	Closely spaced pressure sensors map swallowing, sphincters and gut motor patterns.	Esophageal/anorectal pressure topography and classified motility phenotypes.	Detailed functional localization.	Catheter/position, protocol, normative thresholds and classification do not always predict symptoms/treatment response.
Breath tests	Noninvasive	Exhaled hydrogen, methane, carbon dioxide isotopes or other gases after substrate challenge.	Selected malabsorption, infection or transit proxies.	Low burden and dynamic.	Preparation, microbiome, transit, smoking/exercise and threshold ambiguity; overdiagnosis of bacterial overgrowth.
Liver biopsy and GI histopathology	Invasive	Tissue architecture, inflammation, fibrosis, dysplasia, organisms and molecular markers.	Etiology/stage when tissue changes decisions.	Direct cellular architecture and classification.	Sampling error, observer variability, bleeding/pain and static snapshot.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Microbiome and metabolome profiling	Noninvasive/minimally invasive specimen	Sequencing and mass spectrometry profile organisms, genes and metabolites.	Ecologic/function hypotheses and research subtypes.	Broad view of host-microbe chemistry.	Collection, diet, medication, geography, batch, compositional data and causal ambiguity; commercial reports often lack utility.

10.0.1 History, abdominal and nutritional examination

Measured quantity and transducer. Symptoms, meal/stool/exposure/medication history, inspection, palpation, auscultation, rectal/oral and nutrition/function assessment.

Typical outputs. Urgency, syndrome, localization, alarm features and test strategy.

Strengths. High value and low burden.

Limitations and artifacts. Symptoms are overlapping; examination sensitivity is limited; stigma and recall affect diet/alcohol/stool reports.

Reference and validation strategy. Structured history, validated symptom/stool tools, repeated trajectory and observer training.

Clinical/research maturity. Mature.

Evidence anchors: [6,23]

10.0.2 Blood chemistry, hematology and synthetic markers

Measured quantity and transducer. Enzymatic, immunologic and coagulation assays.

Typical outputs. Anemia/inflammation, cholestatic/hepatocellular pattern, albumin/INR context, electrolytes and organ effects.

Strengths. Widely available and longitudinal.

Limitations and artifacts. Markers are nonspecific and influenced by muscle, nutrition, anticoagulants, hemolysis and acute illness; normal values may coexist with advanced disease.

Reference and validation strategy. Laboratory QC, trends, context and confirmatory testing.

Clinical/research maturity. Mature.

Evidence anchors: [9,37]

10.0.3 Stool microbiology and molecular panels

Measured quantity and transducer. Culture, antigen, microscopy or nucleic-acid detection of pathogens/toxins.

Typical outputs. Evidence of selected infectious agents and resistance markers.

Strengths. Can rapidly identify treatable and outbreak-relevant causes.

Limitations and artifacts. Colonization, prolonged shedding, broad-panel incidental detection and collection/preanalytic errors.

Reference and validation strategy. Clinical syndrome correlation, controls, culture/susceptibility when needed and outcome studies.

Clinical/research maturity. Mature for defined pathogens; panels require stewardship.

Evidence anchors: ^[11,12]

10.0.4 Fecal inflammatory, pancreatic and occult-blood tests

Measured quantity and transducer. Immunochemical or enzymatic markers such as calprotectin, elastase or hemoglobin.

Typical outputs. Intestinal inflammation proxy, exocrine insufficiency proxy or screening signal.

Strengths. Low-burden triage and monitoring for defined claims.

Limitations and artifacts. Medication, dilution, bleeding source, age and pretest probability; not diagnostic alone.

Reference and validation strategy. Assay QC, threshold-by-intended use, repeatability and endoscopic/clinical reference.

Clinical/research maturity. Mature adjunct.

Evidence anchors: ^[6,7]

10.0.5 Upper endoscopy and colonoscopy

Measured quantity and transducer. Flexible video endoscopy with biopsy, polypectomy, hemostasis and therapy.

Typical outputs. Mucosal anatomy, lesions, histology and interventions.

Strengths. Direct visualization and combined diagnosis/treatment.

Limitations and artifacts. Sedation, perforation, bleeding, missed lesions, preparation quality, interval cancer and overuse.

Reference and validation strategy. Equipment/reprocessing QA, preparation, withdrawal/inspection metrics, pathology and complication tracking.

Clinical/research maturity. Mature.

Evidence anchors: [8,39]

10.0.6 Capsule endoscopy and ingestible imaging

Measured quantity and transducer. Swallowed cameras or sensors transmit images/physiology through GI tract.

Typical outputs. Small-bowel mucosa, bleeding source and selected transit/pH/temperature data.

Strengths. Accesses regions beyond standard endoscopy without sedation.

Limitations and artifacts. Retention, incomplete study, localization, image burden, false positives and inability to biopsy/treat.

Reference and validation strategy. Bench battery/transmission, patency selection, blinded comparison and management impact.

Clinical/research maturity. Established for selected small-bowel indications; broader ingestibles emerging.

Evidence anchors: [8,34]

10.0.7 Endoscopic ultrasound

Measured quantity and transducer. High-frequency ultrasound from GI lumen with needle sampling or therapy.

Typical outputs. Pancreatic/biliary/subepithelial lesions, staging and drainage access.

Strengths. High local resolution and tissue acquisition.

Limitations and artifacts. Sedation, bleeding, infection, pancreatitis, sampling error and operator dependence.

Reference and validation strategy. Equipment QA, competency, cytology/pathology reference and complication surveillance.

Clinical/research maturity. Mature specialist method.

Evidence anchors: [8,26]

10.0.8 ERCP and therapeutic biliary/pancreatic endoscopy

Measured quantity and transducer. Cannulation, contrast fluoroscopy, sphincterotomy, extraction, dilation, stenting and sampling.

Typical outputs. Duct diagnosis during therapy, drainage and obstruction treatment.

Strengths. Powerful minimally invasive therapy.

Limitations and artifacts. Pancreatitis, bleeding, infection, perforation and radiation; diagnostic-only use is usually inappropriate when safer imaging exists.

Reference and validation strategy. Appropriateness, fluoroscopy/dose, cannulation/complication metrics and outcome tracking.

Clinical/research maturity. Mature high-risk procedure.

Evidence anchors: [8,39]

10.0.9 Abdominal ultrasound and Doppler

Measured quantity and transducer. Reflected sound and frequency shift image organs, fluid and vessels.

Typical outputs. Gallstones, ducts, liver texture, masses, ascites, portal/hepatic flow and procedure guidance.

Strengths. Portable, inexpensive and no ionizing radiation.

Limitations and artifacts. Body habitus/bowel gas, operator and protocol dependence; texture is non-specific.

Reference and validation strategy. Phantom QA, standardized views, reader reliability and cross-imaging/pathology reference.

Clinical/research maturity. Mature.

Evidence anchors: [9,38]

10.0.10 Elastography

Measured quantity and transducer. Ultrasound or MR measures tissue response to shear waves as a stiffness proxy.

Typical outputs. Liver stiffness and sometimes fat/other tissue properties.

Strengths. Noninvasive fibrosis-risk stratification and longitudinal potential.

Limitations and artifacts. Inflammation, congestion, cholestasis, meals, obesity and platform differences alter stiffness; not direct histology.

Reference and validation strategy. Phantoms, fasting/protocol, reliability metrics and biopsy/outcome validation.

Clinical/research maturity. Mature for defined liver claims.

Evidence anchors: [9,37]

10.0.11 CT, MRI and MRCP

Measured quantity and transducer. Cross-sectional X-ray or magnetic imaging with contrast and duct-focused sequences.

Typical outputs. Inflammation, obstruction, masses, vasculature, staging, complications and bile/pancreatic ducts.

Strengths. Comprehensive anatomy and rapid acute assessment with CT; high soft-tissue/duct contrast with MRI.

Limitations and artifacts. Radiation/contrast, motion, incidental findings, protocol and reconstruction dependence.

Reference and validation strategy. Dose/phantom QA, protocol harmonization, pathology/procedure reference and utility.

Clinical/research maturity. Mature.

Evidence anchors: [8,26,39]

10.0.12 pH-impedance monitoring

Measured quantity and transducer. Electrical impedance and pH detect bolus movement and acid exposure.

Typical outputs. Reflux burden, acid/nonacid events and symptom association.

Strengths. Objective physiology when diagnosis or procedure selection is uncertain.

Limitations and artifacts. Catheter behavior change, day variability, symptom-report bias, thresholds and association statistics.

Reference and validation strategy. Sensor calibration, protocol on/off therapy, repeatability and outcome prediction.

Clinical/research maturity. Mature specialist method.

Evidence anchors: [7,8]

10.0.13 High-resolution manometry

Measured quantity and transducer. Closely spaced pressure sensors map swallowing, sphincters and gut motor patterns.

Typical outputs. Esophageal/anorectal pressure topography and classified motility phenotypes.

Strengths. Detailed functional localization.

Limitations and artifacts. Catheter/position, protocol, normative thresholds and classification do not always predict symptoms/treatment response.

Reference and validation strategy. Pressure calibration, standardized maneuvers, normative data and outcome-linked validation.

Clinical/research maturity. Mature specialist method.

Evidence anchors: [7,8]

10.0.14 Breath tests

Measured quantity and transducer. Exhaled hydrogen, methane, carbon dioxide isotopes or other gases after substrate challenge.

Typical outputs. Selected malabsorption, infection or transit proxies.

Strengths. Low burden and dynamic.

Limitations and artifacts. Preparation, microbiome, transit, smoking/exercise and threshold ambiguity; overdiagnosis of bacterial overgrowth.

Reference and validation strategy. Gas calibration, standardized substrate/protocol and clinical reference.

Clinical/research maturity. Established for narrow indications; overused in some contexts.

Evidence anchors: [6,7]

10.0.15 Liver biopsy and GI histopathology

Measured quantity and transducer. Tissue architecture, inflammation, fibrosis, dysplasia, organisms and molecular markers.

Typical outputs. Etiology/stage when tissue changes decisions.

Strengths. Direct cellular architecture and classification.

Limitations and artifacts. Sampling error, observer variability, bleeding/pain and static snapshot.

Reference and validation strategy. Specimen adequacy, standardized scoring, reader agreement and clinical/molecular integration.

Clinical/research maturity. Mature reference with limitations.

Evidence anchors: [5,9]

10.0.16 Microbiome and metabolome profiling

Measured quantity and transducer. Sequencing and mass spectrometry profile organisms, genes and metabolites.

Typical outputs. Ecologic/function hypotheses and research subtypes.

Strengths. Broad view of host-microbe chemistry.

Limitations and artifacts. Collection, diet, medication, geography, batch, compositional data and causal ambiguity; commercial reports often lack utility.

Reference and validation strategy. Standardized collection, controls, absolute quantification where possible, external replication and intervention trials.

Clinical/research maturity. Research/frontier for most clinical claims.

Evidence anchors: [10,13,43]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise gastrointestinal, hepatobiliary, and pancreatic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

11.0.1 Intended use and measurand

Design requirements. Define the precise gastrointestinal, hepatobiliary, and pancreatic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [34,36]

11.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [36,38]

11.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

Evidence anchors: ^[43,44]

11.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[37]

11.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[37,43]

11.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: ^[42]

11.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[42]

11.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: ^[35,38]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Food safety, sanitation and vaccination	Population prevention	Reduce enteric/hepatitis transmission through safe water/food, hygiene, immunization and infection-control systems.	High-impact public-health evidence; effectiveness depends on infrastructure and coverage.	Unequal access, outbreak disruption and behavioral burden without structural support.	Incidence, coverage, water/food quality, outbreaks and equity.
Nutrition therapy	Behavioral/clinical	Adjust nutrient composition, texture, timing, fiber or exclusion according to mechanism while preserving adequacy and social feasibility.	Effective for defined diseases and symptom phenotypes; expert support reduces deficiency and unnecessary restriction.	Malnutrition, eating disorder, cost, stigma and difficult adherence.	Symptoms, intake, weight/growth, labs, function and reintroduction plan.
Acid suppression and mucosal protection	Pharmacology	Reduce acid or protect mucosa in ulcer, reflux and bleeding-risk contexts.	Strong benefit for defined acid-mediated disease and prophylaxis in selected high-risk patients.	Infection/nutrient associations, rebound, interactions, long unnecessary use and symptom misattribution.	Indication, symptom/healing response, duration, adverse effects and deprescribing opportunities.
Antimicrobial and antiviral therapy	Etiologic pharmacology	Eradicate <i>H. pylori</i> , treat selected enteric/biliary infection and cure/suppress viral hepatitis.	Can cure causal infection and prevent cancer/organ failure; regimen depends on resistance and stage.	Resistance, toxicity, interactions, microbiome disruption, adherence and access.	Test of cure/virologic response, liver function, susceptibility/local ecology and adverse effects.
Anti-inflammatory, immune and biologic therapy	Pharmacology	Suppress or redirect mucosal/hepatic immune pathways in IBD, autoimmune and selected cholestatic disease.	Induces and maintains remission in defined diseases; objective monitoring can guide care.	Infection, malignancy signals, organ toxicity, immunogenicity, cost and reproductive considerations.	Symptoms plus objective inflammation, drug/toxicity markers selectively, infection screening and vaccination.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Motility and gut-brain therapies	Pharmacologic/behavioral	Alter motility, transit, sensation, central modulation or pelvic-floor coordination.	Modest-to-substantial benefit in selected functional/motility phenotypes; matching and expectations matter.	Diarrhea/constipation, cardiac/neurologic effects, dependence on chronic use and stigma.	Symptom and function goals, stool pattern, adverse effects, diet and psychological outcomes.
Endoscopic therapy	Procedure	Remove lesions, stop bleeding, dilate/ablate, drain, stent or treat obesity/reflux in selected contexts.	Can avoid surgery and provide definitive treatment when selection and expertise are appropriate.	Bleeding, perforation, infection, pancreatitis, sedation and repeat-procedure burden.	Technical/clinical success, complications, pathology and durable patient-important outcomes.
Surgery and transplantation	Procedure/organ replacement	Resect disease, correct anatomy, restore continuity or replace liver/intestine in end-stage disease.	Curative/life-saving for selected cancer, inflammation, obstruction and failure.	Perioperative harm, altered nutrition/function, rejection/immunosuppression, recurrence and access.	Complications, pathology, graft, nutrition, function, survival and quality of life.
Metabolic and liver-directed treatment	Multisystem therapy	Treat adiposity, glucose, lipids, alcohol and specific fibrotic/metabolic targets to reduce liver and systemic risk.	Weight and diabetes treatment improves steatosis and some outcomes; disease-specific drugs depend on stage and evidence.	Weight cycling, drug effects, cost, interactions and overreliance on enzymes rather than fibrosis/outcomes.	Weight/metabolic factors, fibrosis risk, liver events, cardiovascular/kidney outcomes and function.
Pancreatic enzyme and nutritional replacement	Replacement therapy	Replace digestive enzymes and fat-soluble/micronutrients when exocrine function is inadequate.	Improves maldigestion, weight and symptoms when dosing/timing are correct.	Pill burden, cost, underdosing, constipation and rare adverse effects.	Weight/stool symptoms, nutrition labs, adherence/timing and diabetes.
Microbiota-directed therapy	Biologic/ecologic intervention	Use defined antibiotics, fecal microbiota products, diet or live biotherapeutics for selected conditions.	Strongest established use is prevention of recurrent <i>C. difficile</i> with regulated products/approaches; other claims remain variable.	Pathogen transmission, resistance genes, unpredictable ecology, donor/product variability and commercialization.	Recurrence, adverse infection, product traceability, resistance and long-term outcomes.
Rehabilitation and palliative digestive care	Supportive care	Address fatigue, nutrition, continence, pain, nausea, pruritus, function, caregiver burden and goals.	Improves quality and supports complex chronic/advanced disease.	Fragmented access, opioid/sedation effects and late referral.	Symptoms, intake, function, participation, caregiver outcomes and goals.

12.0.1 Food safety, sanitation and vaccination

Target and mechanism. Reduce enteric/hepatitis transmission through safe water/food, hygiene, immunization and infection-control systems.

Evidence and expected outcome. High-impact public-health evidence; effectiveness depends on infrastructure and coverage.

Risks, burden, and equity. Unequal access, outbreak disruption and behavioral burden without structural support.

Response monitoring. Incidence, coverage, water/food quality, outbreaks and equity.

Evidence anchors: [28,29]

12.0.2 Nutrition therapy

Target and mechanism. Adjust nutrient composition, texture, timing, fiber or exclusion according to mechanism while preserving adequacy and social feasibility.

Evidence and expected outcome. Effective for defined diseases and symptom phenotypes; expert support reduces deficiency and unnecessary restriction.

Risks, burden, and equity. Malnutrition, eating disorder, cost, stigma and difficult adherence.

Response monitoring. Symptoms, intake, weight/growth, labs, function and reintroduction plan.

Evidence anchors: [6,20]

12.0.3 Acid suppression and mucosal protection

Target and mechanism. Reduce acid or protect mucosa in ulcer, reflux and bleeding-risk contexts.

Evidence and expected outcome. Strong benefit for defined acid-mediated disease and prophylaxis in selected high-risk patients.

Risks, burden, and equity. Infection/nutrient associations, rebound, interactions, long unnecessary use and symptom misattribution.

Response monitoring. Indication, symptom/healing response, duration, adverse effects and deprescribing opportunities.

Evidence anchors: [7,8,40]

12.0.4 Antimicrobial and antiviral therapy

Target and mechanism. Eradicate *H. pylori*, treat selected enteric/biliary infection and cure/suppress viral hepatitis.

Evidence and expected outcome. Can cure causal infection and prevent cancer/organ failure; regimen depends on resistance and stage.

Risks, burden, and equity. Resistance, toxicity, interactions, microbiome disruption, adherence and access.

Response monitoring. Test of cure/virologic response, liver function, susceptibility/local ecology and adverse effects.

Evidence anchors: [7,9,11]

12.0.5 Anti-inflammatory, immune and biologic therapy

Target and mechanism. Suppress or redirect mucosal/hepatic immune pathways in IBD, autoimmune and selected cholestatic disease.

Evidence and expected outcome. Induces and maintains remission in defined diseases; objective monitoring can guide care.

Risks, burden, and equity. Infection, malignancy signals, organ toxicity, immunogenicity, cost and reproductive considerations.

Response monitoring. Symptoms plus objective inflammation, drug/toxicity markers selectively, infection screening and vaccination.

Evidence anchors: [7,30,41]

12.0.6 Motility and gut-brain therapies

Target and mechanism. Alter secretion, transit, sensation, central modulation or pelvic-floor coordination.

Evidence and expected outcome. Modest-to-substantial benefit in selected functional/motility phenotypes; matching and expectations matter.

Risks, burden, and equity. Diarrhea/constipation, cardiac/neurologic effects, dependence on chronic use and stigma.

Response monitoring. Symptom and function goals, stool pattern, adverse effects, diet and psychological outcomes.

Evidence anchors: [7,17,18]

12.0.7 Endoscopic therapy

Target and mechanism. Remove lesions, stop bleeding, dilate/ablate, drain, stent or treat obesity/reflux in selected contexts.

Evidence and expected outcome. Can avoid surgery and provide definitive treatment when selection and expertise are appropriate.

Risks, burden, and equity. Bleeding, perforation, infection, pancreatitis, sedation and repeat-procedure burden.

Response monitoring. Technical/clinical success, complications, pathology and durable patient-important outcomes.

Evidence anchors: [8,39]

12.0.8 Surgery and transplantation

Target and mechanism. Resect disease, correct anatomy, restore continuity or replace liver/intestine in end-stage disease.

Evidence and expected outcome. Curative/life-saving for selected cancer, inflammation, obstruction and failure.

Risks, burden, and equity. Perioperative harm, altered nutrition/function, rejection/immunosuppression, recurrence and access.

Response monitoring. Complications, pathology, graft, nutrition, function, survival and quality of life.

Evidence anchors: [9,30,32]

12.0.9 Metabolic and liver-directed treatment

Target and mechanism. Treat adiposity, glucose, lipids, alcohol and specific fibrotic/metabolic targets to reduce liver and systemic risk.

Evidence and expected outcome. Weight and diabetes treatment improves steatosis and some outcomes; disease-specific drugs depend on stage and evidence.

Risks, burden, and equity. Weight cycling, drug effects, cost, interactions and overreliance on enzymes rather than fibrosis/outcomes.

Response monitoring. Weight/metabolic factors, fibrosis risk, liver events, cardiovascular/kidney outcomes and function.

Evidence anchors: [2,9,14]

12.0.10 Pancreatic enzyme and nutritional replacement

Target and mechanism. Replace digestive enzymes and fat-soluble/micronutrients when exocrine function is inadequate.

Evidence and expected outcome. Improves maldigestion, weight and symptoms when dosing/timing are correct.

Risks, burden, and equity. Pill burden, cost, underdosing, constipation and rare adverse effects.

Response monitoring. Weight/stool symptoms, nutrition labs, adherence/timing and diabetes.

Evidence anchors: [6,40]

12.0.11 Microbiota-directed therapy

Target and mechanism. Use defined antibiotics, fecal microbiota products, diet or live biotherapeutics for selected conditions.

Evidence and expected outcome. Strongest established use is prevention of recurrent *C. difficile* with regulated products/approaches; other claims remain variable.

Risks, burden, and equity. Pathogen transmission, resistance genes, unpredictable ecology, donor/product variability and commercialization.

Response monitoring. Recurrence, adverse infection, product traceability, resistance and long-term outcomes.

Evidence anchors: [10,11,34]

12.0.12 Rehabilitation and palliative digestive care

Target and mechanism. Address fatigue, nutrition, continence, pain, nausea, pruritus, function, caregiver burden and goals.

Evidence and expected outcome. Improves quality and supports complex chronic/advanced disease.

Risks, burden, and equity. Fragmented access, opioid/sedation effects and late referral.

Response monitoring. Symptoms, intake, function, participation, caregiver outcomes and goals.

Evidence anchors: [31,32,33]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can microbiome profiles guide routine personalized diet or therapy?	Microbiota influence host metabolism and immunity, and selected microbiota-directed interventions work in narrow conditions.	Stable individual recommendations that improve outcomes across populations and platforms.	Profiles vary with diet, medication, geography, collection and analysis; association is often mistaken for mechanism.	Multisite randomized strategy trials with standardized methods and predefined clinical outcomes.
How should noninvasive liver fibrosis tests be combined?	Scores and elastography can identify low- and high-risk groups in defined settings.	Optimal sequential algorithms, retesting intervals and thresholds across disease, age and primary care populations.	Spectrum, inflammation, congestion, obesity and platform differences shift performance.	Prospective pathway trials comparing outcomes, referrals, biopsy and cost rather than accuracy alone.
What mechanisms define disorders of gut-brain interaction?	Multiple peripheral and central mechanisms are supported, and treatments can improve symptoms/function.	Actionable mechanistic subtypes with reliable biomarkers.	Symptoms fluctuate and mechanisms overlap; invasive or high-dimensional tests may not transport.	Longitudinal challenge studies followed by mechanism-matched randomized trials.
Can ingestible sensors improve care beyond diaries and intermittent tests?	Capsules can measure images and selected internal variables during transit.	Whether continuous internal data changes treatment and outcomes without retention, privacy or false-positive harm.	Localization, calibration and physiologic interpretation are difficult in a moving, chemically variable tract.	Claim-specific bench models and randomized utility studies in narrowly selected patients.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within gastrointestinal, hepatobiliary, and pancreatic health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

13.0.1 Can microbiome profiles guide routine personalized diet or therapy?

Established knowledge. Microbiota influence host metabolism and immunity, and selected microbiota-directed interventions work in narrow conditions.

Unresolved issue. Stable individual recommendations that improve outcomes across populations and platforms.

Why current evidence is insufficient. Profiles vary with diet, medication, geography, collection and analysis; association is often mistaken for mechanism.

Research that would change confidence. Multisite randomized strategy trials with standardized methods and predefined clinical outcomes.

Evidence anchors: [10,13,42]

13.0.2 How should noninvasive liver fibrosis tests be combined?

Established knowledge. Scores and elastography can identify low- and high-risk groups in defined settings.

Unresolved issue. Optimal sequential algorithms, retesting intervals and thresholds across disease, age and primary care populations.

Why current evidence is insufficient. Spectrum, inflammation, congestion, obesity and platform differences shift performance.

Research that would change confidence. Prospective pathway trials comparing outcomes, referrals, biopsy and cost rather than accuracy alone.

Evidence anchors: [9,45]

13.0.3 What mechanisms define disorders of gut-brain interaction?

Established knowledge. Multiple peripheral and central mechanisms are supported, and treatments can improve symptoms/function.

Unresolved issue. Actionable mechanistic subtypes with reliable biomarkers.

Why current evidence is insufficient. Symptoms fluctuate and mechanisms overlap; invasive or high-dimensional tests may not transport.

Research that would change confidence. Longitudinal challenge studies followed by mechanism-matched randomized trials.

Evidence anchors: [17,18,46]

13.0.4 Can ingestible sensors improve care beyond diaries and intermittent tests?

Established knowledge. Capsules can measure images and selected internal variables during transit.

Unresolved issue. Whether continuous internal data changes treatment and outcomes without retention, privacy or false-positive harm.

Why current evidence is insufficient. Localization, calibration and physiologic interpretation are difficult in a moving, chemically variable tract.

Research that would change confidence. Claim-specific bench models and randomized utility studies in narrowly selected patients.

Evidence anchors: [34,36]

13.0.5 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within gastrointestinal, hepatobiliary, and pancreatic health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: [46]

13.0.6 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: [42]

13.0.7 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: [26,45]

13.0.8 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: [34]

13.0.9 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: [23]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Multimodal symptom-mechanism cohort	Which combinations of motility, reflux, inflammation, diet and central/autonomic state explain disabling GI symptoms?	Prospective repeated-meal cohort with pH/impedance or motility subsets, biomarkers, sleep/stress, diet and functional outcomes; prespecified causal graph.	Replicated subgroups predict response to targeted therapy beyond symptom labels.	Clusters are unstable or driven by site, diet logging and test availability.
0. Ingestible metrology program	Can pH, pressure, gas and temperature capsules remain calibrated through GI conditions?	Bench fluid/pressure/temperature/chemical cycling plus human comparison to catheter/reference methods and localization uncertainty.	Prespecified errors met across transit states and failure states are identifiable.	Average agreement masks drift, retention or state-dependent bias.
0. Liver pathway utility trial	Does staged noninvasive fibrosis assessment reduce decompensation and inequity?	Cluster-randomized primary-care pathway with automatic risk calculation, elastography access and treatment linkage.	More etiologic treatment and fewer liver events without excessive referrals or subgroup gaps.	Detection rises without treatment or outcomes.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for gastrointestinal, hepatobiliary, and pancreatic health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

14.0.1 0. Multimodal symptom-mechanism cohort

Question. Which combinations of motility, reflux, inflammation, diet and central/autonomic state explain disabling GI symptoms?

Design. Prospective repeated-meal cohort with pH/impedance or motility subsets, biomarkers, sleep/stress, diet and functional outcomes; prespecified causal graph.

Success criterion. Replicated subgroups predict response to targeted therapy beyond symptom labels.

Failure or stop criterion. Clusters are unstable or driven by site, diet logging and test availability.

Evidence anchors: [17,42,46]

14.0.2 0. Ingestible metrology program

Question. Can pH, pressure, gas and temperature capsules remain calibrated through GI conditions?

Design. Bench fluid/pressure/temperature/chemical cycling plus human comparison to catheter/reference methods and localization uncertainty.

Success criterion. Prespecified errors met across transit states and failure states are identifiable.

Failure or stop criterion. Average agreement masks drift, retention or state-dependent bias.

Evidence anchors: [36,37]

14.0.3 0. Liver pathway utility trial

Question. Does staged noninvasive fibrosis assessment reduce decompensation and inequity?

Design. Cluster-randomized primary-care pathway with automatic risk calculation, elastography access and treatment linkage.

Success criterion. More etiologic treatment and fewer liver events without excessive referrals or subgroup gaps.

Failure or stop criterion. Detection rises without treatment or outcomes.

Evidence anchors: [9,47,48]

14.0.4 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for gastrointestinal, hepatobiliary, and pancreatic health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: [43,48]

14.0.5 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

Evidence anchors: [46]

14.0.6 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: [36,42]

14.0.7 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: [45,47]

14.0.8 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: [23]

14.0.9 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: [47,48]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03D	Endocrine and Metabolic Health	Nutrient flux, adiposity, diabetes, lipids and endocrine pancreas.	metabolic disease or treatment affects liver, bile or GI function.
03F	Renal and Urinary Health	Fluid/electrolyte consequences, hepatorenal physiology, stones and medication handling.	diarrhea, liver failure or nutrition alters kidney risk.
03H	Immune and Lymphatic Health	Mucosal immunity, IBD, celiac, autoimmune liver and immune therapy.	inflammation or immune suppression is central.
03L	Oral and Craniofacial Health	Chewing, swallowing, oral microbiome and nutrition entry.	upstream oral function changes digestive health.
05	Brain, Cognition, Emotion and Behavior	Gut-brain interaction, pain, eating behavior and substance use.	symptoms and behavior require nervous-system integration.
08	Infection, Immunity, Host-Microbe and Biosecurity	Enteric/hepatitis infection, AMR and microbiome ecology.	pathogens, transmission or stewardship are involved.
11	Population Health and Prevention	Food systems, sanitation, screening and viral elimination.	population infrastructure determines exposure and access.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Cholestasis	Impaired bile formation or flow, recognized by context, biochemical pattern and imaging rather than one analyte.
Enteric nervous system	Intrinsic neural network of the GI tract coordinating motility, secretion and local reflexes.
Maldigestion	Failure to break nutrients into absorbable forms; distinct from mucosal malabsorption.
Portal hypertension	Elevated pressure gradient across the portal venous system, usually from increased hepatic resistance.
Visceral hypersensitivity	Amplified perception or response to internal-organ stimuli, influenced by peripheral and central processes.
Dysbiosis	A descriptive alteration in microbial ecology; it does not by itself establish causation or a treatment target.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.